AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

# T318

# Applied Network Security

## Final Examination
## Makeup-Fall Semester 2024/2025

## Date: 16/Jan./2025

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| **Number of Exam Pages:** 4 (including this cover sheet) | **Time Allowed:** 3 h |

**Instructions:**

- Total Marks: 50
- Total Number of Questions:5
- Marks Distribution:
  - Question 1: 10
  - Question 2: 10
  - Question 3: 10
  - Question 4: 10
  - Question 5: 10
- Write your answer in the answer booklets. Answers given on the exam paper will not be marked.
- The use of electronic devices that could have a memory is NOT permitted.
- Be sure you write your name and ID on the External Answer booklet.



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**Question 1: Answer the following questions [10 Marks]**

1. Assume a password file storing the SHA1 hashes of passwords is on a Linux PC and the file is readable by all users of that PC. A practical approach to make the PC more secure against offline password guessing is:
   A. Automatically disable access to an account if too many incorrect attempts are made.
   B. Limit the speed at which passwords can be entered at the terminal.
   C. Automatically check the passwords when initially created by users and reject the password if it is a "weak" password.
   D. Use the MD5 hash function instead of SHA1 hash function.
2. What does the term "plaintext" refer to?
   A. Encrypted data
   B. Original, unencrypted data
   C. A type of cipher
   D. A key used in encryption
3. In public key cryptography, which key is used to encrypt the data?
   A. Public key
   B. Private key
   C. Session key
   D. Hash key
4. If you are developing a database that stores login credentials for users (e.g. username and password), you should:
   A. Save the password as plaintext
   B. Calculate the hash of the password and save the hash value
   C. Calculate the MAC of the password using a secret key, and save the MAC
   D. Encrypt the password with AES and save the ciphertext
5. Which of the following is not a valid form of attack on an encryption system?
   A. Brute force attack
   B. Ciphertext-only attack
   C. Man-in-the-middle attack
   D. Hashing attack
6. Which protocol is commonly used to secure communications over the Internet?
   A. FTP
   B. HTTP
   C. SSL/TLS
   D. Telnet
7. What is the main function of a digital signature?
   A. To encrypt the message
   B. To verify the sender's identity
   C. To compress the data
   D. To generate random keys
8. Panita's X.509 certificate (which is signed by the certificate authority Pongpak) contains:
   A. Only Panita's public key (and no other keys)
   B. Only Panita's private key (and no other keys)
   C. Panita's public key and Pongpak's public key
   D. Panita's private key and Pongpak's public key
   E. Panita's public key and Pongpak's private key
   F. Panita's private key and Pongpak's private key
9. In the context of network security, what does the term "firewall" refer to?
   A. A type of encryption
   B. A device that monitors and controls incoming and outgoing network traffic
   C. A software application for email security
   D. A method of securing passwords

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10. What is the main purpose of the Certificate Authority (CA)?

A. To generate encryption keys
B. To issue digital certificates
C. To encrypt data
D. To decrypt data

**Question 2: Answer the following questions [10 Marks]**

- In Diffie-Hellman key exchange, each user chooses a private $X$, calculates a public $Y$ and after exchanging $Y$, calculate $K$ where for user $A$:
  a) $Y_a = \alpha^{Xa} \bmod q$
  b) $K_a = Y_b^{Xa} \bmod q$
  c) $q$ is a prime number, $X_a < q$, and $\alpha$ is a primitive root of $q$
- You and other users have agreed to use the public values q = 11 and α = 2. You just received a message from Steve containing his public value, Yb = 9. You choose a private value Xa = 8.
  d) What public value will you send to Steve? Show calculations. (4 marks)
  e) What secret will you share with Steve? Show calculations. (4 marks)
- Assume Tark has intercepted the packets used in the Diffie-Hellman key exchange between you and Steve.
  f) Show the steps Tark takes to determine Steve's private value. (2 marks)

**Question 3: Answer the following questions [10 Marks]**

a) What is the primary purpose of a KDC in a cryptographic system?(3 Marks)

- Given the plaintext {000102030405060708090A0B0C0D0E0F} and the key {01010101010101010101010101010101} (7 Marks)

b) Show the original contents of State, displayed as a 4 x 4 matrix.(2 Marks)
c) Show the value of State after initial AddRoundKey. (2 Marks)
d) Show the value of State after SubBytes. (2 Marks)
e) Show the value of State after ShiftRows. (1 Marks)

**Question 4: Answer the following questions [10 Marks]**

- Consider the mechanism illustrated in Figure 1.

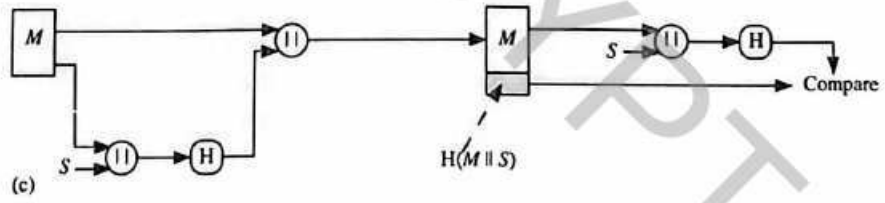



Figure 1: Security mechanism 1

a) What is a security service that this mechanism provides? (1 mark)
b) Explain (or define) the one-way property (also called pre-image resistant property) of a hash function. (1.5 marks)
c) Explain how an attacker can defeat the above security service if the function H() did not have the one-way property. (1.5 marks)

- Consider the mechanism illustrated in Figure 2

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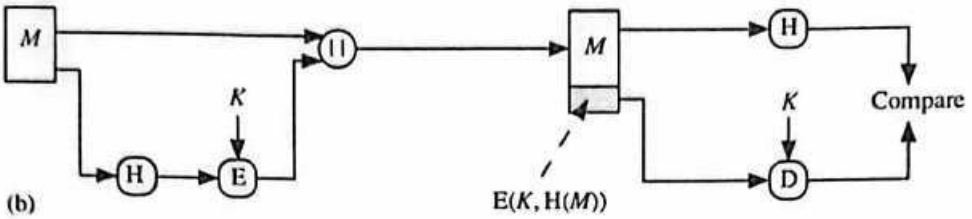
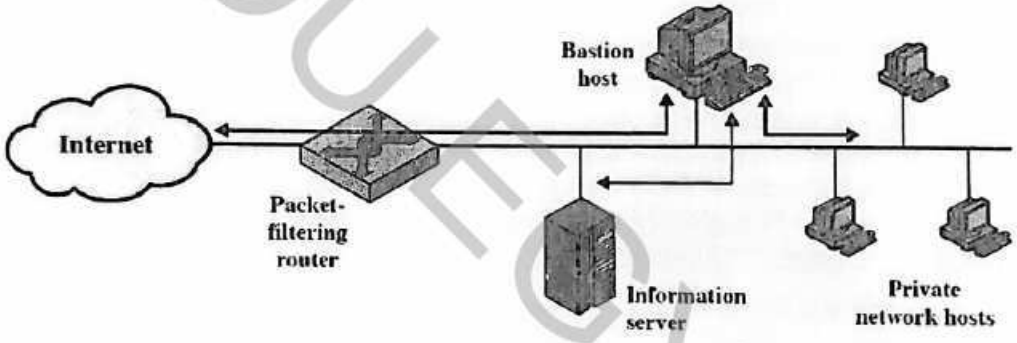



Figure 2: Security mechanism 2

d) What is a security service that this mechanism provides? (1 mark)
e) Explain (or define) the weak collision resistant property (also called second preimage resistant property) of a hash function. (1.5 marks)
f) Explain how an attacker can defeat the above security service if the function H() did not have the weak collision resistant property. (1.5 marks)
g) What is the difference between a hash function and a MAC function? (2 marks)

**Question 5: Answer the following questions [10 Marks]**

A single homed bastion host is one way to configure a network using both a packet filter firewall/router, as well as an application level firewall (bastion host).




a) Explain the difference between a packet filter firewall/router and an application level firewall. Include an advantage of each type of firewall. (4 marks)
b) Describe two rules that must be configured on the packet filter router in this configuration. (3 marks)
c) What is an advantage of using this configuration (as opposed to using only an application-level firewall)? (2 marks)
d) What is the difference between this configuration and a dual homed bastion host configuration? Explain how the dual homed configuration provides an advantage over the single homed configuration. (2 marks)

End of Questions

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